



GymGenius

GYM

AlignAI

**PERSONAL TRAINING,
PERFECTED BY AI**



A presentation by team GymGenius





Problem Statement

Revolutionize workout safety and performance with AI-driven corrections that optimize exercise form and elevate results. Instant, intelligent feedback guides users towards perfect posture in every session, every rep, every goal.



TRAINING

THE AlignAI SOLUTION



AI POSE ESTIMATION

Deep learning extracts precise joint positions in real time for accurate movement tracking and analysis.



INTELLIGENT FEEDBACK

Dynamic overlays and audio alerts provide instant, data-driven corrections for each user and exercise.



PERFORMANCE ANALYTICS

Algorithms assess joint angles and detect posture errors against expert-approved standards.

OUR METHOD

AlignAI's FLOW

INPUT CAPTURE

Camera streams user movements in real time.



DETECTION

AI pinpoints joint positions and body angles.



CORRECTION

System highlights posture errors, giving instant feedback.



REFINEMENT

Immediate adjustment enabling optimal form.





KEY FEATURES



REAL-TIME FLAGS

AlignAI configures itself using just your device camera—no wearables or extra hardware needed.



AUTOMATED SETUP

Mistakes flagged by audio beeps and visual overlays, including skeleton and joint highlights.



EXPERT VALIDATION

Form rules are designed and verified by medical experts to ensure reliable, safe movement guidance.

A black and white photograph of two women in a gym, performing a kettlebell squat. The woman in the foreground is in a deep squat position, holding a kettlebell with both hands. She is wearing a dark sports bra, leggings, and sneakers. The woman in the background is also in a squat position, holding a kettlebell. The gym has large windows, racks of kettlebells, and other exercise equipment. The text "POSTURE PRECISION. POWERED BY INTELLIGENCE." is overlaid on the right side of the image in white and orange.

**POSTURE
PRECISION.
POWERED BY
INTELLIGENCE.**

TECHNOLOGY **STACK**

01

FRONTEND

Built with HTML, CSS, and JavaScript for smooth browser-based interaction.

03

FEEDBACK LAYER

Uses HTML Canvas for instant overlays and Web Audio API for alerts.

02

POSE ESTIMATION

Live movement tracking via TensorFlow.js (MoveNet) and MediaPipe BlazePose.

04

ANALYTICS

Tracks reps, form accuracy, and session time using LocalStorage.



PROOF & PROGRESS

Prepare

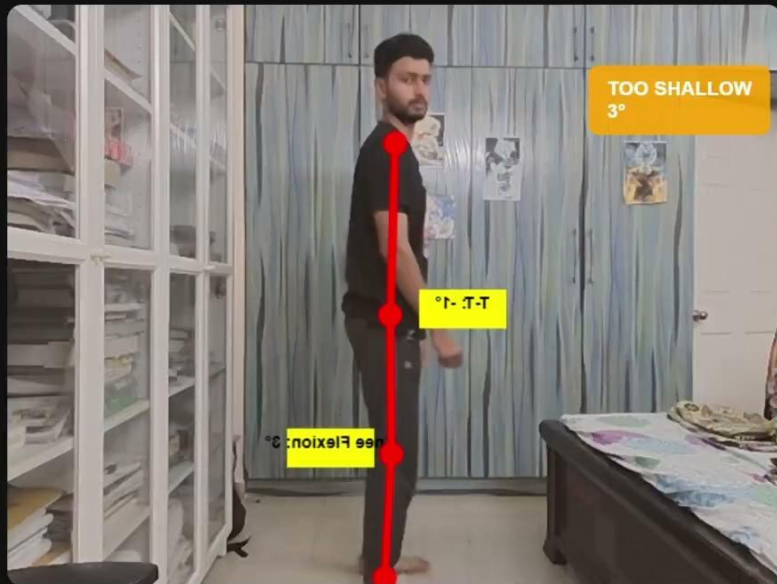


Perform



Perfect





AlignAI: Squat Form Coach

Squat (USC Biomechanics) ▾

Stop Exercise

Exercise: SQUAT (USC Biomechanics Standards)

- Stand sideways to camera (left side visible)
- Feet 100-150% shoulder width apart, toes ~30° outward
- Target: Medium depth (90-110° knee flexion)
- Maintain neutral spine throughout movement
- Device: Place camera at hip height ~3-5 feet from you

⚠ Form Corrections:

- Squat deeper - reach medium depth

SQUAT DEEPER

Current: 3°

Target: 90-110°

OUR GOALS

IMPACT

- Breaks down barriers to expert posture guidance—no trainer required.
- Delivers instant, professional feedback anywhere, anytime.
- Minimizes injury risk while building long-term form habits.
- Drives consistent progress with motivational insights and analytics.



VISION

- Expand to more exercises and personalized training programs.
- Implement adaptive feedback powered by historical session data.
- Launch mobile support and a standalone application.
- Collaborate with physiotherapists for even smarter model refinement.

The resistance that you fight
physically in the gym and the
resistance that you fight in life
can only build a strong character.

—Arnold **Schwarzenegger**





GymGenius

THANK YOU!

Our impact—quantified, qualified, undeniable.

-Team GymGenius

Surampally Poojitha, Shraddha Desaraju, Rohan Reddy, Rohit Kothari,
Neelam Chandra Kiran

